2023-2024
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People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
National Higher School of Mathematics

Resit exam ($1^{h}30$)

Name :	Groupe:

Exercice 1. We give in the following statistical table :

class	frequency	Rel. Freq	Cum. Rel. Freq
$[20, 30[$	100	0.125	0.125
$[30, 40[$	140	0.175	0.3
$[40, x_1[$	125	0.156 25	0.456 25
$[x_1, 70[$	200	0.25	0.706 25
$[70, 100[$	180	0.225	0.931 25
$[100, x_2[$	55	0.068 75	1

1. Given that the median of this distribution is equal to 56.8, calculate x_1.

The median $\widetilde{X} \in [x_1, 70[$ and given by

$$\widetilde{X} = L_1 + \left(\frac{\frac{n}{2} - Cf_b}{f_m}\right) \times w = x_1 + \left(\frac{\frac{800}{2} - 365}{200}\right) \times (70 - x_1) = 56.8$$

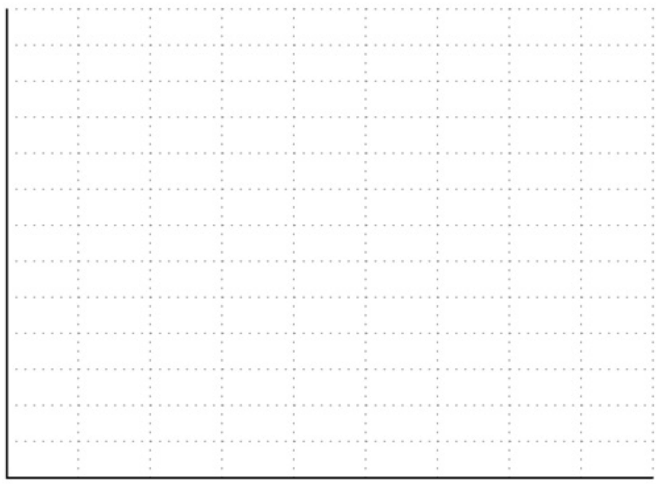
therefore, $x_1 = 54$.

2. Given that the arithmetic mean of the population studied is equal to 60.5, calculate the bound x_2.

$$\overline{X} = 60.5 = 25 \times 0.125 + 35 \times 0.175 + 47 \times 0.15625 + 62 \times 0.25 + 85 \times 0.225 + \frac{100 + x_2}{2} \times 0.068\,75$$

which gives $x_2 = 170$

3. Plot the histogram of the distribution and calculate its mode.



Clearly, The highest frequency is for the fourth class $[54, 70[$ to which the mode $\widehat{X}$ belongs and it's given by

$$\widehat{X} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) \times w = 54 + \left(\frac{(200-125)}{(200-125)+(200-180)}\right) \times (70-54) = 66.632$$

Exercice 2. The following table gives the distribution of 100 people questioned, according to two characteristics X and Y : X is the monthly income expressed in thousands of dinars and Y is the number of vacation departures during a given year

$X \backslash Y$	0	1	2	3	4	$\sum_X$
$[10, 15[$	4	4	6	1	0	15.0
$[15, 20[$	6	7	5	2	0	20.0
$[20, 25[$	4	4	6	5	1	20.0
$[25, 30[$	0	5	7	6	2	20.0
$[30, 35[$	1	8	6	5	5	25.0
$\sum_Y$	15	28	30	19	8	100.0

1. Give the number of people receiving a salary of at least 20 thousand dinars.

   At least 20 thousand dinars $\Longleftrightarrow X \in [20, 25[\cup[25, 30[\cup[30, 35[\Longrightarrow 20 + 20 + 25 = 65.$

2. Give the number of people who have gone on vacation at least twice.

   gone on vacation at least twice $\Longleftrightarrow Y \in \{2, 3, 4\} \Longrightarrow 30 + 19 + 8 = 57.$

3. Among people earning between 20,000 and 25,000 dinars, what is the percentage of those who have gone on vacation at least once?

   The percentage is $\frac{16}{20} \times 100\% = 80\%$

4. Give the marginal distribution of X. Determine the mode and the first quartile.

X	**freq.**	**cum.freq**
$[10, 15[$	15.0	15
$[15, 20[$	20.0	35
$[20, 25[$	20.0	55
$[25, 30[$	20.0	75
$[30, 35[$	25.0	100

the mode $\widehat{X} \in [30, 35[$ and

$$\widehat{X} = 30 + \left(\frac{(100-70)}{(100-70)+(100-0)}\right) \times 5 = 31.154$$

The first quartile $\in [15, 20[$

$$Q_1 = 15 + \left(\frac{\frac{100}{2} - 15}{20}\right) \times 5 = 23.75$$

5. Determine the conditional distribution of Y given that $X \in [15, 20[$, then calculate the conditional mean of the number of vacation departures.

$Y \mid X \in [15, 20[$	0	1	2	3	4	$\sum$
freq.	6	7	5	2	0	20.0

the conditional mean is

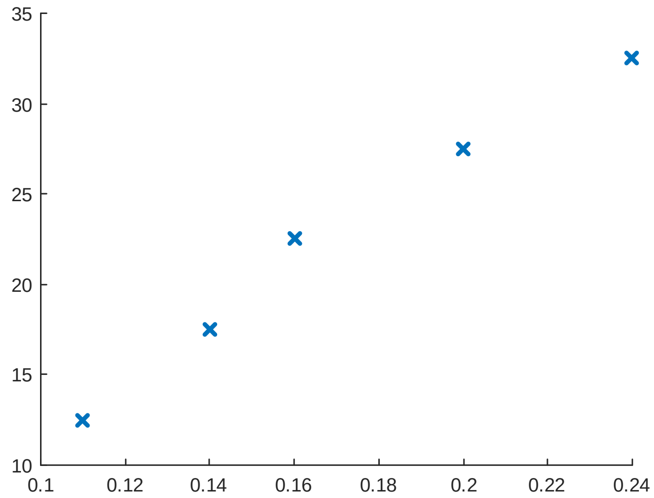
$$\frac{1}{20}(0 \times 6 + 1 \times 7 + 2 \times 5 + 3 \times 2 + 4 \times 0) = 1.15$$

6. define Z ="The holiday departure rate" as the ratio of the number of people who have gone on holiday at least once to the total number of people in the population considered.

(a) Complete the following table:

x_i	12.5	17.5	22.5	27.5	32.5
z_i	0.11	0.14	0.16	0.2	0.24

(b) Draw the scatter plot. What could we conclude?



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(c) Calculate $Cov(X, Z)$ then $\rho(X, Z)$

$$Cov(X, Z) = 0.4$$
$$\rho(X, Z) = 0.9923$$

(d) Write the regression equation of Z on X. What would be the vacation departure rate for people earning 37,500 DA?

$$Z_i = aX_i + b + \varepsilon_i$$
$$\widehat{a} = 0.0064$$
$$\widehat{b} = 0.026$$

the vacation departure rate for people earning 37,500 DA is estimated by the regression equation of Z on X, thus

$$Z = \widehat{a} \times 37.5 + \widehat{b} = 0.0064 \times 37.5 + 0.026 = 0.266$$